

Lecture Summarization with Fine-tuned T5

How does fine-tuning a general model on academic content change its summarization behavior?

Stef Elrington

<https://github.com/stan5k/lecture-summarizer>

General Assembly: Advanced Applications of AI capstone

05/07/2026

Why is lecture summarization different from other abstract summarization?

Most training data is the news

State-of-the-art summarizers (BART, T5) are pre-trained primarily on news articles.

Lectures differ greatly from the news

Longer, denser, more conceptual; transcripts are spoken text, often noisy.

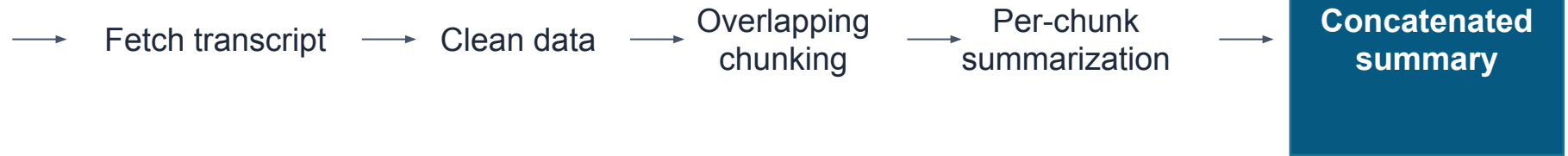
The capstone project

Whether fine-tuning on academic data helps or hurts off the shelf models in summarizing lectures

Pipeline overview



URL



The three model configurations

Two models used as baseline to offer reference for final model performance

BART-large-CNN

Trained on CNN/DailyMail news

T5-base

Multi-task mixture of unsupervised and supervised tasks

The final model

T5-base fine-tuned

arxiv-summarization
(~4,300 examples)

Test set of lectures deliberately diverse

Machine Learning

STEM, technical

But what is a neural network?

3Blue1Brown · 18:39 · 3,357 words

History

Humanities, argumentative

Neoliberalism and the End of History

MIT lecture · 15:06 · 2,057 words

Psychology

Social science, narrative

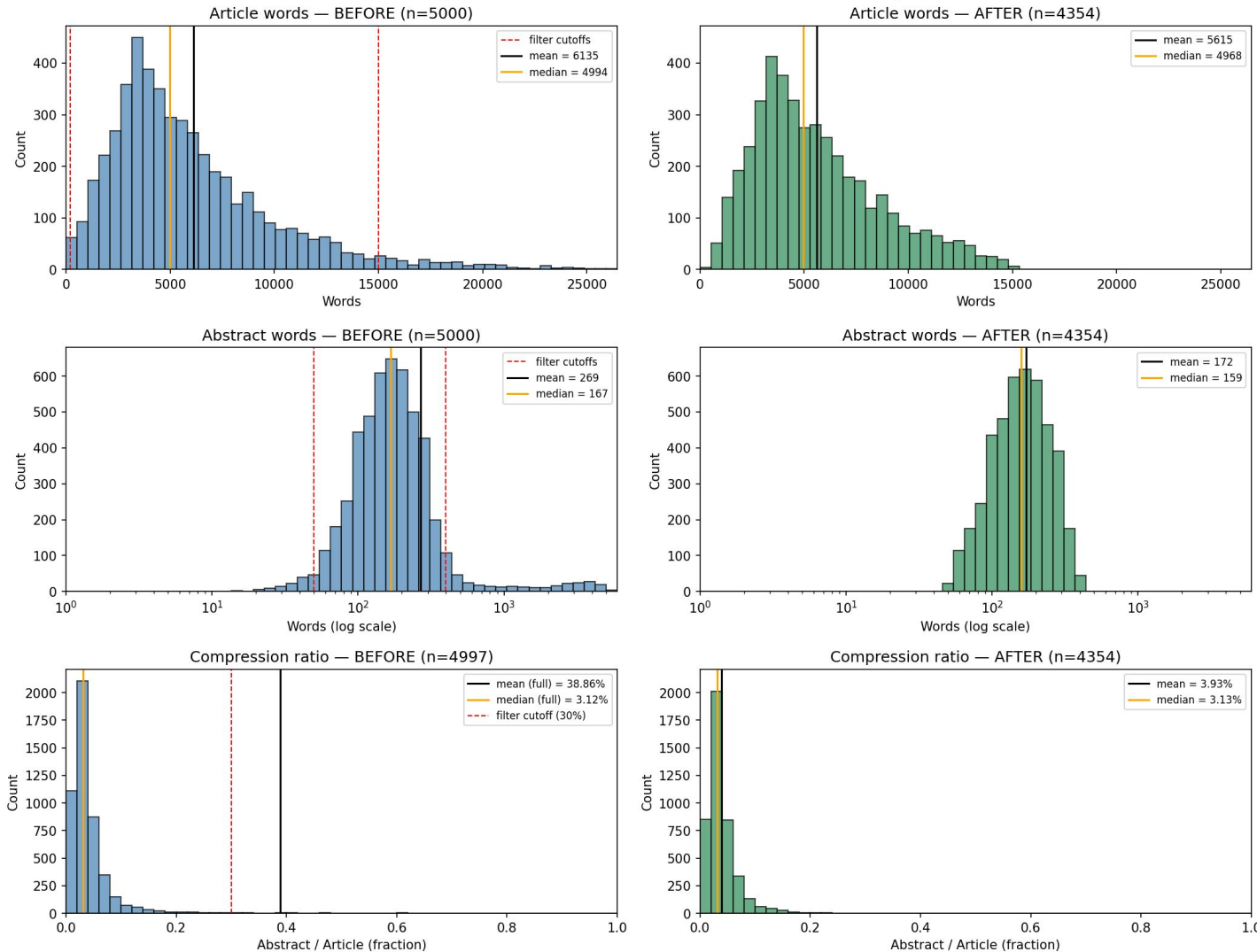
Why Having More Doesn't Make You Happier

Yale Online · 26:34 · 5,857 words

arxiv-summarization data skews closer in similarity to Machine Learning

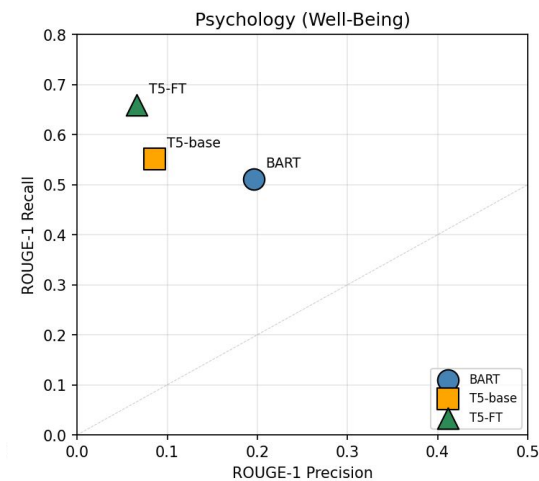
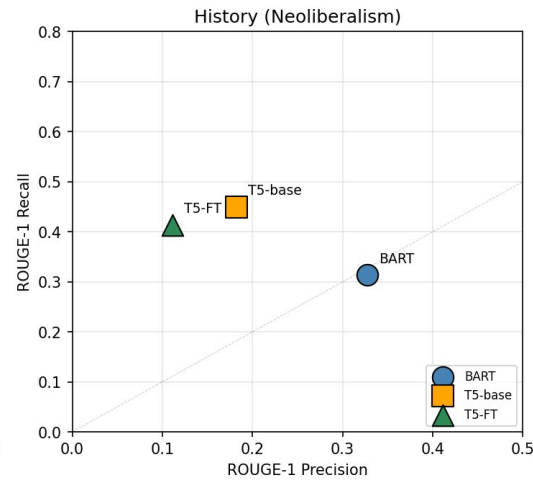
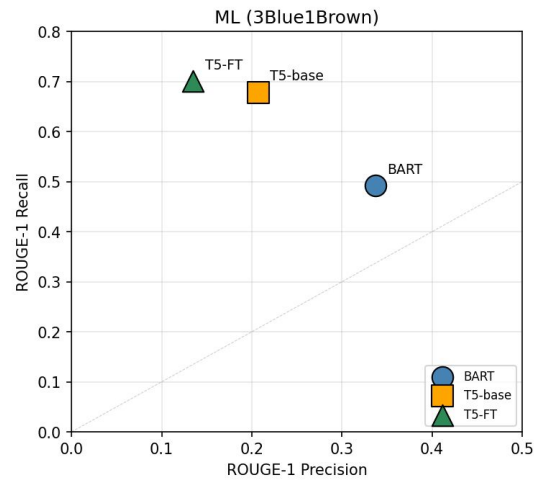
Cleaning the arxiv training data

Training subset distributions: before vs after filtering



- 5,000 random arxiv-summarization examples → 4,354 after filtering
- Outliers included zero-word "articles" paired with full abstracts (compression > 4000%), so filtered on:
 - article 200–15K words
 - abstract 50–400 words
 - compression ≤ 30%

The resulting trade-off on test data conciseness vs coverage



BART

Highest precision

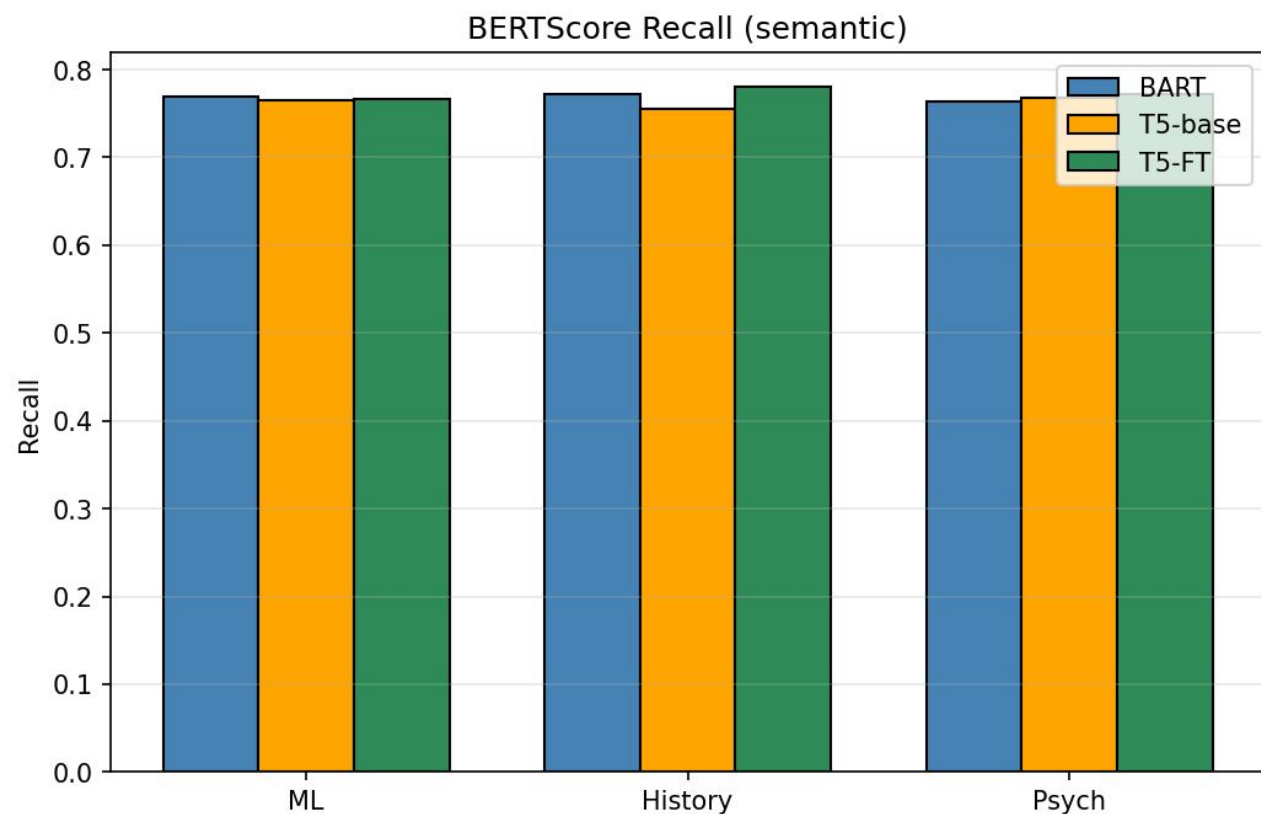
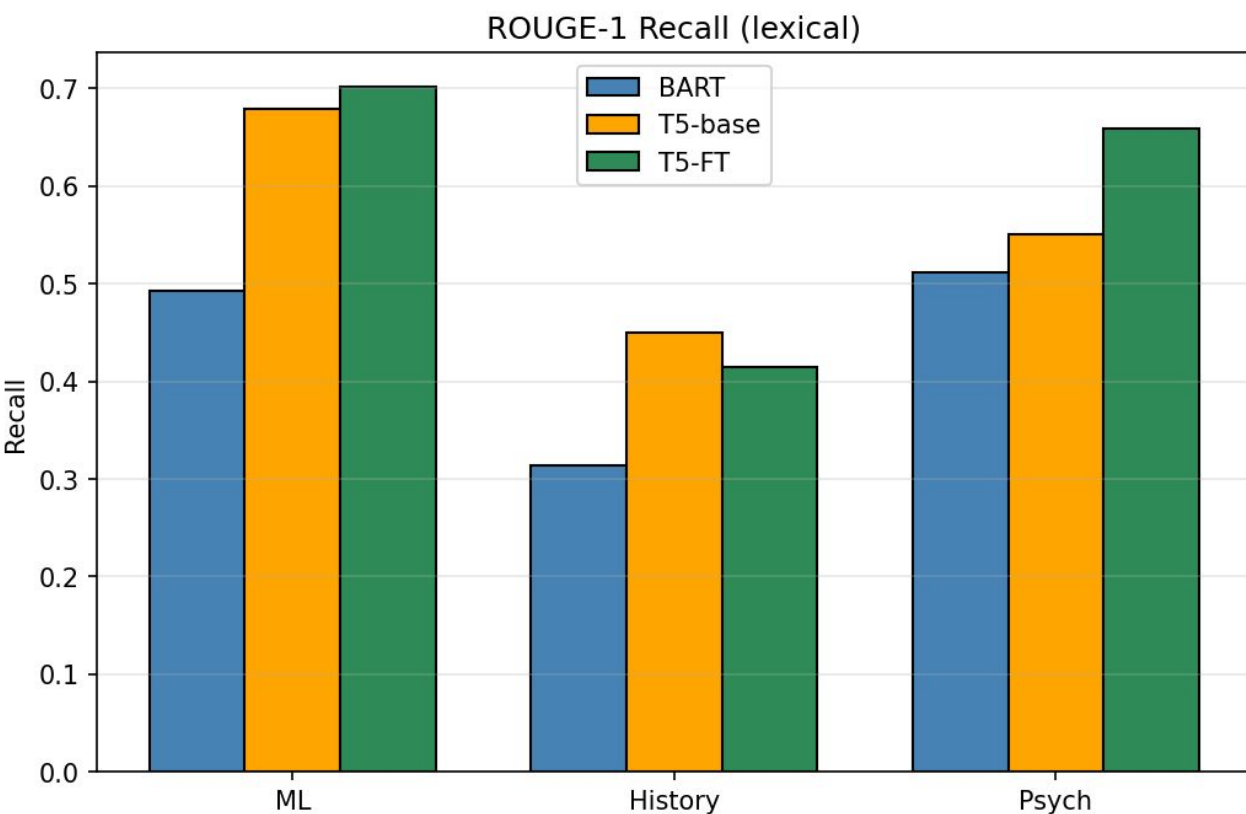
Fine-tuned T5

Highest recall but at the expense of being verbose

T5 baseline

Weakest overall

Fine-tuned T5 covering more reference content on 2 of 3 via Rouge-1 and BERT



Some of the observed failures in model outputs

Failure mode	BART	T5-base	T5-fine-tuned
	Clean precise prose with no weird artifacts	Summaries read like they were news articles	Summaries were verbose and it did not perform so well with History lecture
Hallucinated attribution	—	Yes ("John Avlon" on ML)	—
Repetition (4-grams)	minor	minor	Yes (especially in history)
Boilerplate fixation	—	—	Yes (especially in history)
Length inflation	—	moderate	severe

Live demo

github.com/stan5k/lecture-summarizer

Limitations and future work

What I learnt

- Fine-tuned T5 is verbose and repetitive. It also lacks organizational structure
- Light fine-tuning (3 epochs, 4.3K examples) of this model can be optimized. Including the number of epochs, and even the quality of the T5 prompt (instead of just saying “summarize:”)
- 3 lectures is relatively small for an evaluation set
- Naive concatenation of chunk summaries led to visible repetition (between adjacent chunks)

Future work

- Larger, mixed-domain fine-tuning corpus (humanities + STEM)
- Hierarchical summarization (summarize the summaries)
- Larger lecture evaluation set with human-written references
- Investigate domain mismatch failure modes more systematically